Practical No. 13 – F-Ratio Test for Equality of Variances and t-Test for Correlation Coefficient

Blue = given data (inputs). Black = formulas. Each test is solved by the critical-value method and the p-value method; both must agree.

Ex.1 Silver plating – F-test for variances (left-tailed)

H0: $\sigma_1^2 = \sigma_2^2$ vs H1: $\sigma_1^2 < \sigma_2^2$ ($F = s_1^2/s_2^2$, df = n1−1, n2−1)

Sample SD s1 (Company 1)	0.035	
Sample size n1	12	
Sample SD s2 (Company 2)	0.062	
Sample size n2	12	
Significance level α	0.05	
df1 = n1 − 1	11	
df2 = n2 − 1	11	
Test statistic F	**0.3187**	$F = s_1^2/s_2^2$
Critical value (lower) F(1−α)	0.3549	FINV(1−α, df1, df2)
p-value (left tail)	0.03531	1 − FDIST(F, df1, df2)
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Company 1's silver plating has significantly less variability than Company 2's.

Ex.2 Oxide thickness with two gases – F-test for variances (two-tailed)

H0: $\sigma_1^2 = \sigma_2^2$ vs H1: $\sigma_1^2 \neq \sigma_2^2$ (is either gas preferable?)

Sample SD s1 (angstroms)	1.96	
Sample size n1	16	
Sample SD s2 (angstroms)	2.13	
Sample size n2	16	
Significance level α	0.05	
df1 = n1 − 1	15	
df2 = n2 − 1	15	
Test statistic F	**0.8467**	$F = s_1^2/s_2^2$
Lower critical value F(1−α/2)	0.3494	FINV(1−α/2, df1, df2)
Upper critical value F(α/2)	2.8621	FINV(α/2, df1, df2)
p-value (two-tailed)	0.75151	2 × smaller tail area
Decision (critical value method)	**Fail to reject H0**	
Decision (p-value method)	**Fail to reject H0**	

Conclusion: No strong evidence that either gas gives a smaller variance of oxide thickness.

Ex.3 Age and price of equipment – test of ρ = −0.40 (left-tailed)

H0: ρ = −0.40 vs H1: ρ < −0.40 (ρ0 ≠ 0, so Fisher z-transformation is used; the t-test is valid only for ρ0 = 0)

Sample correlation r	-0.58	
Sample size n	25	
Hypothesised ρ0	-0.40	
Significance level α	0.05	
Fisher z of r	-0.6625	$z_r = \frac{1}{2}\ln((1+r)/(1-r))$
Fisher z of ρ0	-0.4236	$z_\rho = \frac{1}{2}\ln((1+\rho 0)/(1-\rho 0))$
Test statistic z	**-1.1201**	$z = (z_r - z_\rho)\sqrt{(n-3)}$
Critical value −z(α)	-1.6449	NORMSINV(α)
p-value (left tail)	0.13133	Φ(z)
Decision (critical value method)	**Fail to reject H0**	
Decision (p-value method)	**Fail to reject H0**	

Conclusion: Not enough evidence that ρ is less than −0.40.

Ex.4 Bivariate normal sample – t-test for correlation (two-tailed)

H0: ρ = 0 vs H1: ρ ≠ 0 ($t = r\sqrt{(n-2)}/\sqrt{(1-r^2)}$, df = n−2)

Sample correlation r	0.50	
Sample size n	15	
Significance level α	0.05	
Degrees of freedom	13	df = n − 2
Test statistic t	**2.0817**	$t = r\sqrt{(n-2)}/\sqrt{(1-r^2)}$
Critical value ±t(α/2, df)	2.1604	TINV(α, df)
p-value (two-tailed)	0.05770	TDIST(|t|, df, 2)
Decision (critical value method)	**Fail to reject H0**	
Decision (p-value method)	**Fail to reject H0**	

Conclusion: Sample data does not indicate the existence of correlation in the population.